

VAISHNAVI SAMALA

Hyderabad, Telangana

📞 741-623-5675

✉ samalavaish77@gmail.com

🌐 [linkedin.com/in/](https://www.linkedin.com/in/)

🐙 github.com/

Education

Geethanjali College of Engineering and Technology

Sep. 2022 – March 2026

Bachelor of Technology

Hyderabad, Telangana

Relevant Coursework

- Database Management Systems
- Statistics & Probability
- Data Structures
- Network & Data Security
- Artificial Intelligence
- Data Mining
- Operating Systems
- Python Programming

Experience

Supraja Technologies

May 2025 – August 2025

Cybersecurity Intern

Hyderabad, Telangana

- Collaborated in a team of 4 to develop a Python-based steganography tool that encodes and conceals text within image files and transmits them securely via SMTP email.
- Enhanced the tool with key features including File Hide, Image Preview, and an intuitive GUI, improving overall usability and adoption.
- Integrated a user database to manage credentials and usage data, gaining hands-on experience with data storage and user management systems.
- Actively contributed across the full development lifecycle — brainstorming, testing, debugging, and iterative improvement — working collaboratively using version control.

Projects

AI-Driven Cloud Security Threat Predictor | *Python, Flask, React.js, MySQL, ML*

2026

- Built a cloud security system using ML models (Random Forest, XGBoost, Isolation Forest) to detect anomalous user behaviour in real time.
- Applied behavioural drift analysis and risk scoring (low/medium/high) to prioritize threats, reducing analyst response time.
- Integrated Explainable AI (XAI) to surface interpretable anomaly explanations, improving analyst decision-making accuracy.
- Developed role-based dashboards with real-time alerts to support security incident monitoring and response workflows.

Sales Performance Analysis | *SQL (PostgreSQL), CTEs, Window Functions*

2025

- Loaded and queried the Northwind relational database (14 tables, 830+ orders) writing 5 business-focused SQL queries using joins, CTEs, and window functions.
- Used LAG() for month-over-month revenue trends and RANK() to identify top products; found top product earned \$141K despite lower volume than rank-3 product.
- Identified 11 dormant customers using date-math CTEs (most at-risk: 657 days since last order) and calculated 4.57% late-shipment rate using conditional aggregation.

Customer Churn & Retention Analysis | *Python, Pandas, Tableau Public*

2025

- Cleaned a 7,043-row Telco churn dataset in Python/Pandas: fixed TotalCharges stored as text, standardized 7 inconsistent categorical columns, and engineered a numeric ChurnFlag.
- Built a 4-chart interactive Tableau Public dashboard revealing month-to-month customers churn at ~43% vs. ~3% for two-year contracts — strongest churn predictor in dataset.
- Found churned customers have half the average tenure of retained customers (18 vs. 37 months) and pay ~\$13/month more, indicating price sensitivity as a key retention risk.

Technical Skills

Languages & Querying: Python, SQL

Web Development: HTML, CSS, JavaScript, React.js

Visualization & BI: Tableau Public, Power BI, Excel

Cloud & Data Tools: Git/GitHub, Jupyter Notebook

Developer Tools: VS Code, Linux, MySQL, Flask

Core Concepts: Data Cleaning, Exploratory Data Analysis, Data Visualization, KPI Reporting, A/B Testing